R V COLLEGE OF ENGINEERING

(An autonomous institution affiliated to VTU, Belgaum)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

IM233AI – WORK SYSTEMS DESIGN

UNIT I

Q No	Question	Marks	Difficulty Level	BT Level	COs
1.	Define the Pyramidal Structure of Work.	2	L	1	1
2.	Define a Work System as a physical entity.	2	L	1	1
3.	How is Work Systems defined as a field of professional practice?	2	L	1	1
4.	Distinguish between a Task and a Work Element	2	L	2	1
5.	What is the primary objective of Work Measurement?	2	L	1	1
6.	Provide the basic formula for Productivity and define its terms.	2	L	1	1
7.	List two common ways to improve productivity in a manufacturing system.	2	L	2	1
8.	What is a Manual Work system?	2	L	1	1
9.	What are Allowances in the context of standard time? Give one example.	2	L	2	1
10.	What is meant by a Worker-Machine System?	2	L	1	1
11.	Give two examples of an operation carried out using a Worker-Machine System.	2	L	2	1
12.	Define an Automated Work System.	2	M	1	1
13.	What is the main difference between a Worker-Machine System and an Automated System regarding the role of the worker?	2	M	2	1
14.	What is the formula for Normal Time in time study? Define the terms.	2	M	1	1
15.	In a worker-machine system, define the Machine Cycle Time.	2	L	1	1
16.	For a single worker operating a single machine, what determines the Cycle Time?	2	M	2	1
17.	State the relationship between Work Methods and Work Measurement	2	M	2	1
18.	What is a Therblig? Who proposed this concept?	2	M	2	1
19.	In the context of the pyramidal structure, what are Basic Motion Elements and why are they important?	2	M	2	1
20.	What is Observed Time and mention how it is obtained in a time study.	2	M	2	1